

# Multi-Threshold Pass-Count Preservation under Bounded Score Perturbations

Research Architecture Runtime

operator/multi-threshold

2026-07-29

## Abstract

For a finite threshold list, the pass-count  $C(x) = \sum_i \mathbf{1}\{x \geq T_i\}$  is invariant under  $|x' - x| \leq \varepsilon$  whenever every cut is buffered ( $x \geq T_i + \varepsilon$  or  $x < T_i - \varepsilon$ ). Lean `LEAN_FULLL`.

This theorem is: Derived

## 1 Problem

The multi-threshold operator returns the Nat pass-count  $C(x; T_\bullet) = |\{i : x \geq T_i\}|$  over a finite list of cuts (equality passes).

## 2 Stability notion

Perturbation:  $|x' - x| \leq \varepsilon$ . Hypothesis `allCoordsStable`: each  $T_i$  satisfies  $x \geq T_i + \varepsilon$  or  $x < T_i - \varepsilon$ . Stability:  $C(x'; T_\bullet) = C(x; T_\bullet)$ . Sharpness: if  $x$  lies in  $[T - \varepsilon, T + \varepsilon)$  for a single cut, some admissible  $x'$  changes the count.

## 3 Definitions

**Definition 1** (Pass bit / count).  $\text{passBit}(x, T) = 1$  if  $x \geq T$  else 0;  $C(x; []) = 0$  and  $C(x; T::\text{tail}) = \text{passBit}(x, T) + C(x; \text{tail})$ .

**Assumptions.**  $\varepsilon \geq 0$ ,  $|x' - x| \leq \varepsilon$ , and `allCoordsStable`.

## 4 Theorem

**Theorem 2** (Count preservation). *If  $|x' - x| \leq \varepsilon$  and every  $T \in T_\bullet$  is buffered as above, then  $C(x'; T_\bullet) = C(x; T_\bullet)$ .*

**Theorem 3** (Sharpness). *If  $T - \varepsilon \leq x < T + \varepsilon$ , there exists  $|x' - x| \leq \varepsilon$  with  $C(x'; [T]) \neq C(x; [T])$ .*

## 5 Intuition

Each buffered cut is a scalar threshold bit; the count is the sum of those bits. If any cut sits in its unstable band, that bit can flip.

## 6 Examples

**Example 4.**  $T = (1, 3, 5)$ ,  $x = 4$ ,  $\varepsilon = 0.2$ : passes  $T = 1, 3$  so  $C = 2$ ; buffered cuts keep  $C = 2$ . If  $T = 3$ ,  $x = 3.1$ ,  $\varepsilon = 0.2$ , then  $x' = x - \varepsilon$  can drop the pass bit.

## 7 Proof

**Preservation.** Induct on the threshold list. The empty list has count 0. For  $T::\text{tail}$ , `allCoordsStable` gives a scalar buffer on  $T$ , so threshold preservation yields  $\text{passBit}(x', T) = \text{passBit}(x, T)$ . The inductive hypothesis equates the tail counts. Adding proves Theorem 2.

**Sharpness.** Assume  $T - \varepsilon \leq x < T + \varepsilon$ . If  $x \geq T$ , take  $x' = x - \varepsilon$  (bit drops). If  $x < T$ , take  $x' = x + \varepsilon$  (bit rises). This proves Theorem 3.

## 8 Formal statement

Lean module

`Research.Operators.MultiThreshold.Preservation`

Source file

`lean/Research/Operators/MultiThreshold/Preservation.lean`

Theorems

- `multi\_threshold\_preservation`
- `multi\_threshold\_sharpness`

**Note**

Depends on `Research.Operators.Threshold.Preservation` per cut.

**Certificate**

`lean/certificates/multi-threshold/multi-threshold-count-preservation`

**Status**

`LEAN_FULL` on Mathlib  $\mathbb{R}$  (`REAL_MATHLIB`)

## 9 Proof dependencies

- `Threshold.Preservation` (each pass bit).
- List induction on the threshold spine.
- No ranking / argmax theorems required.

## 10 Consequences

Unordered pass-counts for multiple cuts; constraint conjunction/disjunction reuse buffered bits. Ordered bucket indices are a separate ranking/bucket operator.